

HAMMERFEST, Norway

WMO: 010520

Lat: 70.683N

Lon: 23.683E

Elev: 266

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.2	10.8	-2.0	5.0	10.1	2.2	6.2	12.2	38.2	34.5	33.7	32.2	7.9	110	0.793

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	69.8	58.5	66.3	56.8	62.8	54.8	59.6	67.9	57.4	65.1	55.3	62.0	8.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.4	66.0	64.5	53.2	60.9	60.9	51.4	56.8	58.5	26.3	67.9	24.8	65.7	23.5	62.0	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.5	26.0	22.5	2.8	75.8	3.2	5.4	0.5	79.7	-1.4	82.8	-3.1	85.9	-5.4	89.8
			1.2	63.3	3.0	3.9	-0.9	66.1	-2.7	68.4	-4.4	70.6	-6.6	73.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	25.0	24.2	26.9	31.9	39.2	46.1	52.5	51.0	46.2	37.6	30.6	28.5	(6)
(7)		DBStd	11.65	7.20	7.38	6.46	5.66	6.23	5.93	6.58	5.87	5.58	6.11	5.77	6.35	(7)
(8)		HDD50	5126	774	723	715	544	343	149	44	57	141	386	583	667	(8)
(9)		HDD65	10329	1239	1143	1180	994	801	566	392	435	565	849	1033	1132	(9)
(10)		CDD50	277	0	0	0	0	6	33	122	89	26	1	0	0	(10)
(11)		CDD65	7	0	0	0	0	0	1	5	1	0	0	0	0	(11)
(12)		CDH74	38	0	0	0	0	0	1	31	6	0	0	0	0	(12)
(13)		CDH80	3	0	0	0	0	0	0	2	1	0	0	0	0	(13)
(14)	Wind	WSAvg	10.0	10.8	10.9	11.6	9.8	9.9	9.5	8.6	8.4	9.0	10.2	9.9	10.9	(14)
(15)	Precipitation	PrecAvg	28.0	2.7	2.2	2.5	1.9	1.7	1.8	2.0	2.1	2.5	3.2	2.6	2.9	(15)
(16)		PrecMax	32.6	5.6	3.8	4.8	3.5	3.9	4.1	5.4	5.2	4.4	5.2	4.7	4.5	(16)
(17)		PrecMin	23.1	0.1	0.4	0.6	0.6	0.4	0.7	0.2	1.0	1.1	1.3	0.3	0.5	(17)
(18)		PrecStd	2.7	1.5	0.8	1.1	0.8	0.8	0.8	1.1	0.9	0.8	0.9	1.3	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.4	41.3	43.0	49.9	63.1	71.5	78.3	71.9	64.3	55.0	46.2	43.4	(19)
(20)			MCWB	39.2	37.6	38.0	42.3	51.4	57.3	65.0	58.8	56.4	48.9	43.1	40.1	(20)
(21)		2%	DB	39.5	39.0	40.5	45.0	56.7	64.8	71.5	67.7	60.4	51.5	43.2	41.3	(21)
(22)			MCWB	36.6	36.3	36.7	40.0	47.9	54.2	60.4	57.7	53.2	46.4	40.4	38.2	(22)
(23)		5%	DB	37.5	36.8	37.8	41.5	52.0	60.1	68.1	64.2	57.3	48.2	40.8	39.3	(23)
(24)			MCWB	34.8	34.3	34.5	37.8	44.9	51.6	58.2	55.8	51.1	44.7	38.1	36.4	(24)
(25)		10%	DB	35.4	34.1	35.8	39.3	48.2	55.7	64.5	60.7	54.5	46.2	38.7	37.3	(25)
(26)			MCWB	32.8	31.7	32.7	36.1	42.8	49.1	56.3	54.0	49.5	43.5	36.1	34.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.5	38.6	38.7	43.1	52.0	59.3	65.7	61.2	56.5	49.1	43.5	40.9	(27)
(28)			MCDWB	41.3	40.3	41.2	48.7	60.5	69.0	75.4	68.6	62.7	54.0	45.4	43.1	(28)
(29)		2%	WB	37.0	36.3	37.0	40.4	48.2	55.0	61.2	58.3	53.8	46.9	40.7	38.5	(29)
(30)			MCDWB	38.8	38.3	39.9	44.3	56.0	63.2	70.0	66.1	59.1	50.4	42.8	40.8	(30)
(31)		5%	WB	34.9	34.3	34.9	38.2	45.5	52.0	58.6	56.1	51.7	45.0	38.1	36.5	(31)
(32)			MCDWB	37.1	36.3	37.5	41.3	51.0	58.9	66.8	62.8	56.4	47.5	40.1	38.8	(32)
(33)		10%	WB	32.7	32.0	32.9	36.4	43.1	49.6	56.5	54.3	49.9	43.4	36.1	34.6	(33)
(34)			MCDWB	34.9	33.7	35.2	39.0	47.3	54.9	63.9	60.1	53.9	45.8	38.3	36.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	8.4	7.8	7.8	8.3	8.9	10.1	8.9	7.4	6.2	7.0	7.7	(35)
(36)			MCDBR	8.3	9.0	8.0	9.0	14.4	16.8	17.5	15.1	11.2	7.8	7.4	8.0	(36)
(37)			MCWBR	7.8	8.2	6.7	6.6	8.2	9.4	9.2	8.2	6.8	6.0	6.9	7.1	(37)
(38)		5% WB	MCDBR	8.3	8.9	7.8	8.6	13.4	15.7	16.7	14.0	10.4	7.0	7.4	8.3	(38)
(39)			MCWBR	8.2	8.5	7.0	6.5	8.2	9.2	9.2	8.3	6.8	5.9	7.2	7.8	(39)
(40)		Clear-Sky Solar Irradiance	taub	N/A	0.258	0.314	0.332	0.345	0.340	0.349	0.335	0.312	0.262	N/A	N/A	(40)
(41)	taud		N/A	2.055	2.154	2.260	2.380	2.483	2.491	2.514	2.514	2.159	N/A	N/A	(41)	
(42)	Ebn at Noon		0	131	209	251	264	270	263	252	224	128	0	0	(42)	
(43)	Edh at Noon		0	12	24	30	30	28	27	23	17	10	0	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0	67	361	922	1390	1525	1375	940	488	133	0	0	(44)	
(45)		RadStd	0	10	43	88	118	141	147	85	54	16	0	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page